# Quiz

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## Main notions learned in this chapter

2/5 points (graded)

Most binary strings of size 1024 can be compressed by 10 bits.

- True
- False (selected)

**Answer**
**Correct:**
**Only one thousandth of all strings of size 1024 can be compressed by 10 bits.**

In Huffman codes, code lengths are inversely proportional to probability.

- True (selected)
- False

**Answer**
**Incorrect:**
**No. Code lengths are proportional to the logarithm of the inverse of the probability.**

In typical texts, the 10th most frequent word is 10 times more frequent than the 100th most frequent word.

- True (selected)
- False

**Answer**
**Correct: True. That's Zipf's law.**

The joint complexity $K(w_1, w_2)$ between the semantics of two words $w_1$ and $w_2$ can be approximated by $-\log(g(w_1 + w_2))$, where $g$ counts the number of occurrence in a search engine.

- True (selected)
- False

**Answer**
**Incorrect:**
**One also needs to know the frequency of the two words, as given by $g(w_1)$ and $g(w_2)$.**

The normalized "Google" distance (NGD) can be in principle as large as $log(N)/2$, where $N$ is the number of pages indexed by the search engine.

- True (selected)
- False

✖

**Answer**
**Incorrect:  No! The NGD is between 0 and 1 (hence the name "normalized").**

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## Problem 1

0/2 points (graded)

We start from a string consisting of one thousand alternations of 0 and 1: `'01' * 1000` . Then we apply the hash function `sha-256` to get a 256-bit long sequence (64 hexadecimal characters).

```
$ python
>>> from hashlib import sha256
>>> S = sha256(('01' * 1000).encode('utf-8')).hexdigest()
>>> print(S)
6f8464fabe54add258fbbf3b85197859808564ad65baf71101b151e19c3e3309
```

What is the best constraint about the complexity of `S` ?

- ◯ It is less than 2000 bits
- ◯ It is less than 1000 bits
- ◉ It is less than 512 bits
- ◯ It is less than 30 bits

✖

**Answer**
**Incorrect:**
**The complexity of the hashed string is only apparent. It cannot be more complex than the initial (periodic) binary string (let's say about 20 bits to encode the binary string as the expression '01' * 1000: about 2 bits for '01'; the remainder is used to encode the operation '*' and its (round) operand '1000') + the complexity of signalling that the hash function was used (we may allow up to 10 additional bits to select the function from the Python library). So 50 bits is a conservative estimate.**

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✖ Incorrect (0/2 points)

## Problem 2

2/2 points (graded)

A 20-bit long pattern is found to occur at locations 1670 and 2100 of a binary string. Do we spare information by encoding the repeat instead of merely leaving the bits intact?

- ◉ Yes, we spare information.
- ◯ No, it's shorter to leave the bits intact.

**Answer**
**Correct:**
**Encoding the repeat requires coding for the pattern length and for the distance between its occurrences. This requires about 5 bits + 9 bits. This is less than 20 (there is still room for a few signal bits).**

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Answers are displayed within the problem

## Problem 3

2/2 points (graded)

The word 'knife' (as a noun) is found 17120 times in a large corpus. This makes it the $2254^{th}$ most frequent word. The word 'cup' is more frequent: it is found 61009 times. Which do you think is the most plausible rank of 'cup' in the list of words sorted by decreasing frequency?

- ( ) 440
- ( ) 990
- (•) 780
- ( ) 1210

**Answer**
**Correct:**
**Zipf's law makes us expect a rank that is about: 17120 * 2254 // 61009 = 632. So 780 is the most likely answer (and actually the correct one).**

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Answers are displayed within the problem

## Problem 4

2.0/2.0 points (graded)

Using "Bing" as Web search engine, we found:

```
kimono 18 600 000 hits
sushi 54 300 000 hits
kimono+sushi 424 000 hits
```

Which value do we get for `NGD(kimono, sushi)`, if we suppose that Bing crawled $8 \times 10^9$ pages?

- (•) 0.8
- ( ) 0.6
- ( ) 0.7

0.5

Answer
Correct:

NGD(kimono, sushi) = (log(54300000,2) - log(424000,2)) / (log(8000000000,2) - log(18600000,2))
= 0.800

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Answers are displayed within the problem

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